

# House Prices

## Section 1 — Loading & Initial Exploration

1. Load train.csv into a pandas DataFrame. Display the first 10 rows, shape, and column names.
  2. Display the data types of all columns. How many columns are numerical and how many are categorical?
  3. Display the statistical summary using describe(). Identify the top 3 numerical columns with the highest standard deviation.
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## Section 2 — Missing Value Treatment

4. Find the percentage of missing values for each column. Display only those columns where missing values exceed 5%.
  5. Drop all columns where more than 40% of values are missing. Print the list of dropped columns and the new shape of the dataframe.
  6. For remaining numerical columns with missing values, fill them with the median. For categorical columns with missing values, fill them with the mode.
  7. Confirm that no missing values remain in the dataset after treatment.
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## Section 3 — Encoding Categorical Variables

8. Identify all categorical columns remaining in the dataset after dropping high-null columns.
  9. Apply Ordinal Encoding to columns that have a clear order such as ExterQual, BsmtQual, KitchenQual, and HeatingQC (Poor < Fair < Typical < Good < Excellent).
  10. Apply One-Hot Encoding to nominal categorical columns such as Neighborhood, SaleType, and MSZoning. Drop the first dummy to avoid multicollinearity.
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## Section 4 — Feature Engineering

11. Create a new column TotalSF = TotalBsmtSF + 1stFlrSF + 2ndFlrSF.
  12. Create a new column HouseAge = YrSold - YearBuilt.
  13. Create a new column IsRemodelled — value should be 1 if YearRemodAdd is different from YearBuilt, otherwise 0.
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## Section 5 — Outlier Detection & Target Variable Analysis

14. Plot a box plot for GrLivArea. Use the IQR method to detect and remove outliers from this column.
  15. Plot the distribution of SalePrice using a histogram. Describe the shape — is it skewed?
  16. Apply log transformation to SalePrice and plot the distribution again. Comment on the difference.
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## Section 6 — Feature Scaling

17. Separate the features (X) and target (y). Use the log-transformed SalePrice as y.
  18. Apply StandardScaler to all numerical columns in X. Verify by printing the mean and standard deviation of any two scaled columns.
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## Section 7 — Train Test Split

19. Split the data into training and test sets using an 80/20 ratio with random\_state=42. Print the shape of all four splits — X\_train, X\_test, y\_train, y\_test.
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## Section 8 — Model Building & Prediction

20. Train a Linear Regression model on the training data. Predict SalePrice on the test set.
  21. Train a Decision Tree Regressor with max\_depth=5 on the same training data. Predict on the test set.
  22. Train a Random Forest Regressor with 100 estimators on the training data. Predict on the test set.
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## Section 9 — Model Evaluation

23. For all three models, calculate and display MAE, RMSE, and R-squared on the test set in a single comparison table.
  24. Plot Actual vs Predicted values for the Random Forest model as a scatter plot. Draw the ideal prediction line ( $y = x$ ) in red.
  25. Plot the Top 10 Feature Importances from the Random Forest model as a horizontal bar chart sorted from highest to lowest.
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## Section 10 — Conclusion

26. Based on the evaluation metrics, which model performed best? Write your answer in a markdown cell with justification.

27. Name two steps you would take to further improve the best model's performance.